Siratul Islam

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Education

Shahjalal University of Science and Technology **Aug. 2023 – Aug 2027**
Bachelor of Science (Hons) in Physics *Sylhet, Bangladesh*

Experience

The Zephyr RTOS Project (Linux Foundation) **Remote**
Biometrics, Authentication, and FIDO2 Subsystems Maintainer *Feb. 2026 – Present*

- Author and Maintainer of Zephyr's Biometrics driver class (PR #100139)
- Author and Maintainer of Zephyr's FIDO2 & Authentication subsystems (PR #104327)
- Review community contributions on various subsystems
- Technologies: Embedded C, Subsystem Architecture, Device Drivers, Device Tree, USB, PSA Crypto

Contributor (Triage) *Jun. 2025 – Present*

- Zephyr Contributor with Triage access and 20+ merged PRs across subsystems, drivers, fixes, and BSPs
- Develop device drivers and board support packages serving embedded developers globally
- Review community contributions and provide technical guidance on driver architecture and device tree bindings

The Linux Kernel **Remote**
Driver Maintainer & Reviewer, Industrial I/O (IIO) Subsystem *Mar. 2026 – Present*

- **Maintained Drivers:** ST VL53L1X (Time-of-Flight sensor), QST QMC5883L (Magnetometer)
- Authored the drivers and Device Tree bindings and used latest Kernel APIs
- Review patches in linux-iio on lore.kernel.org, and collaborate with contributors

Shahjalal University of Science and Technology **Jun. 2025 – Present**
Research Assistant - Department of Electrical and Electronic Engineering *Sylhet, Bangladesh*

- Developed a Zephyr based smart notification/timer/clock system for EEE department with OTA and Firmware signing
- Developing IoT-enabled smart control systems and radar-based occupancy detection for government energy optimization
- Building RFID-based attendance management systems targeting rural educational institutions in Bangladesh

Hackules Inc. **Feb. 2024 – Jun. 2025**
Software Engineer *Remote*

- Led full-stack/frontend development for educational platforms (Opedemy, Teachers Today) serving 1000+ active users

Selected Projects

Wireless Smart Display System | *Zephyr, ESP32, OTA, Display, MQTT, MCUBoot* **source code**

- Wrote the official Zephyr HUB12 Display driver.
- Implemented OTA, including progress indicator and used MCUBoot with firmware signing for enhanced security
- Implemented multiple switchable clock-faces and advanced scheduling features for notifications and timer.

STM32 BME280 HAL Driver | *C, STM32, I2C, Bare-Metal* **source code**

- Developed a custom Hardware Abstraction Layer (HAL) driver in C for the Bosch BME280 environmental sensor
- Implemented reliable I2C communication protocols to parse temperature, humidity, and pressure telemetry

Publications

1. (Under Preparation) S. Islam, M. Rasedujjaman, "BASE: A Vendor-Agnostic Biometric Driver Class for Embedded Real-Time Systems", Target - *IEEE Transactions on Industrial Informatics.*

2. (Under Preparation) S. Islam, M. Rasedujjaman, "Embedding FIDO2 in an Industrial Edge Node: Footprint, Latency, and Scheduling Trade-offs", Target - *IEEE Transactions on Industrial Informatics.*

Technical Skills

Specializations : Zephyr RTOS, Linux Kernel, FIDO2/CTAP2, Embedded Biometrics, IoT Security & Hardware Auth
Embedded Systems: Zephyr RTOS, FreeRTOS, Linux Kernel, STM32 HAL, ESP-IDF, Embedded Linux, ROS2
Platforms & Protocols : STM32, ESP32, nRF52, Raspberry Pi, MQTT, BLE, Wi-Fi, LoRa, USB, NFC, I2C, SPI, UART
Development Tools: Embedded C/C++, Rust, STM32CubeIDE, CMake, PlatformIO, OpenOCD, KiCAD, Git
Full-Stack Web: C#, TypeScript, .NET, SvelteKit, Next.js, PostgreSQL, Docker, Podman